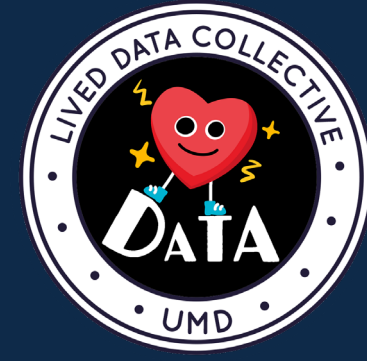


Bringing Data to Life: Designing Data Characters for the Emotional Self

Diego Abarcar Calugay, Isabella Amador, Keke Wu

University of Maryland - College Park

CONTACT: diegoac@terpmail.umd.edu, iamador1@terpmail.umd.edu, kekewu@umd.edu



How can Data Characters represent affective experiences? What visual encodings emerge through customization?

1 INTRODUCTION

Journaling supports emotional reflection, expression, and regulation. However, interpreting these long-term **emotional patterns** can be difficult, as visualizations often fail to grasp **affective nuance** [4].



Figure 1: Word clouds focus on frequency but omit context [4].

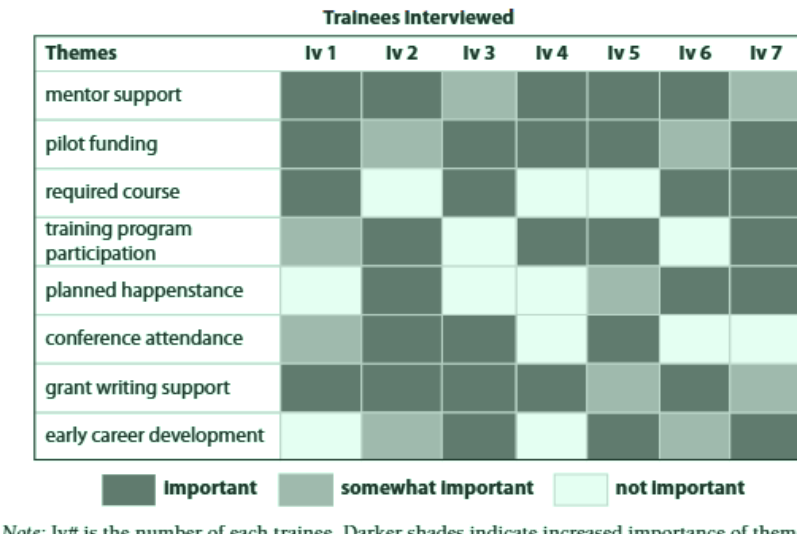


Figure 2: Discrete themes limit nuanced insights [4].

4 PRELIMINARY WALKTHROUGHS

We invited two participants to represent an emotional state with Data Characters. This process demonstrated the **intuitiveness** and **feasibility** of the interface.

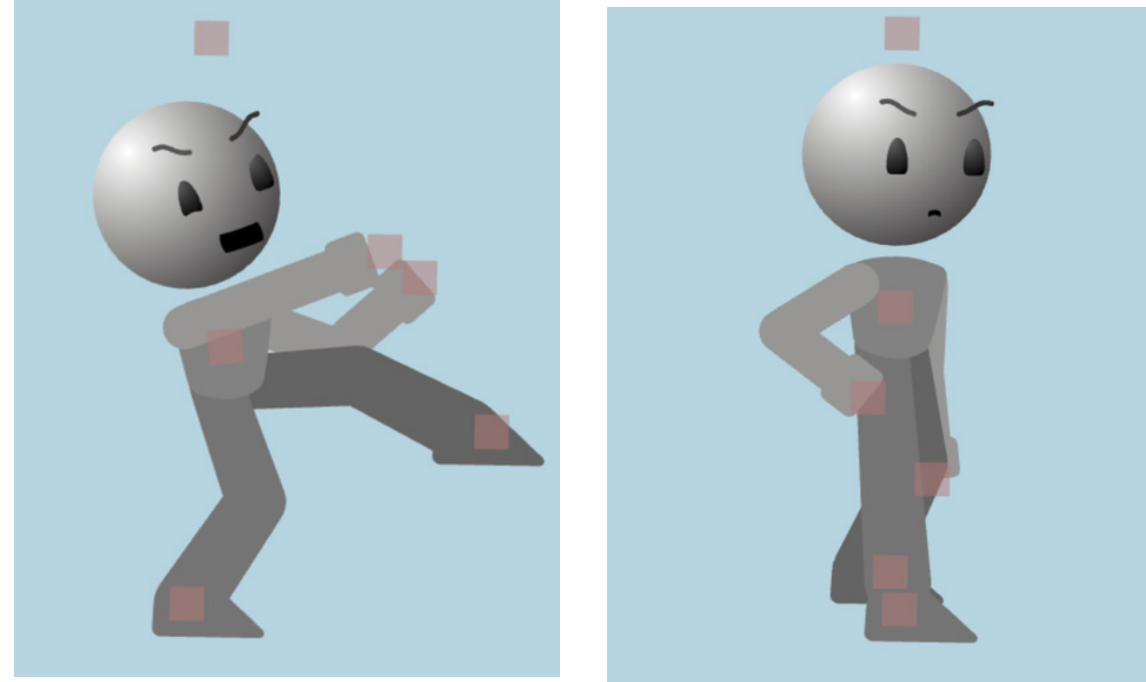


Figure 4: Users generated personalized visualizations of "anger."

2 DESIGN RATIONALE

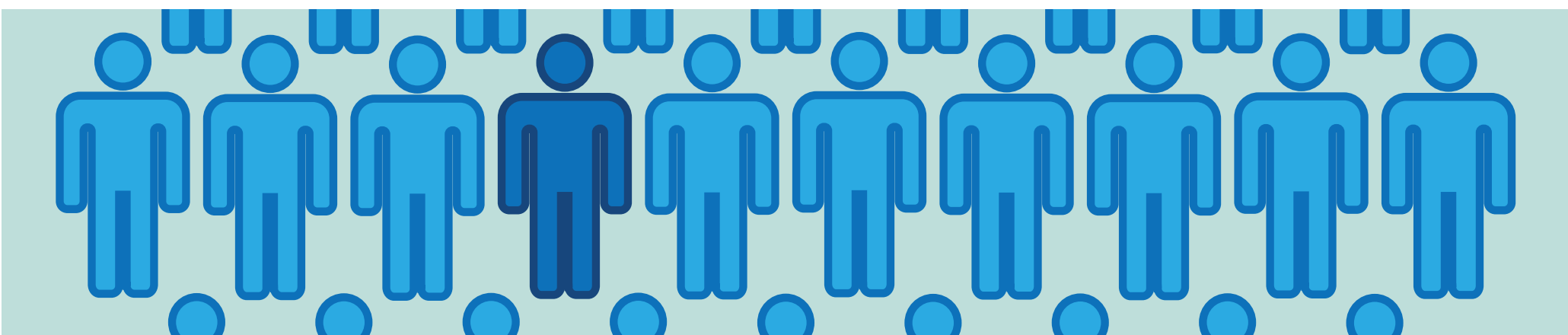


Figure 3: Anthropographics imply the usefulness of human-like visuals for emotional interactions [6].

We sought a visualization that directly conveys affective experiences. We drew from two ideas:

Nonverbal Communication

People naturally express emotions with familiar cues involving facial features and body language.

Anthropomorphism

People assign human traits to nonhuman entities. In visualization, anthropomorphic designs may support **emotional expression and representation**.

- Voices and faces can clarify complex ideas [2].
- Flexible displays that mirror human postures are perceived as emotionally expressive [5].

We propose Data Characters, anthropomorphic representations of emotional content.

3 DESIGN PROBE

We implemented a **customizable** Data Character as a **design probe** [3], allowing users to express affective data by modifying facial and bodily features.

- Sliders offer continuous emotional representation.
- Drag-and-drop handles control limb movement.

6 LIMITATIONS

Nonverbal expressions **may not be universal**, limiting the generalizability of users' designs. Moreover, Data Characters may capture emotions but **not emotional context**, omitting details from rich affective experiences.

7 FUTURE WORK

Overall, we aim to conduct a **larger design-probe study** that explores how Data Characters represent **varied emotional experiences**, revealing common **encoding strategies** for affect.

We hope to...

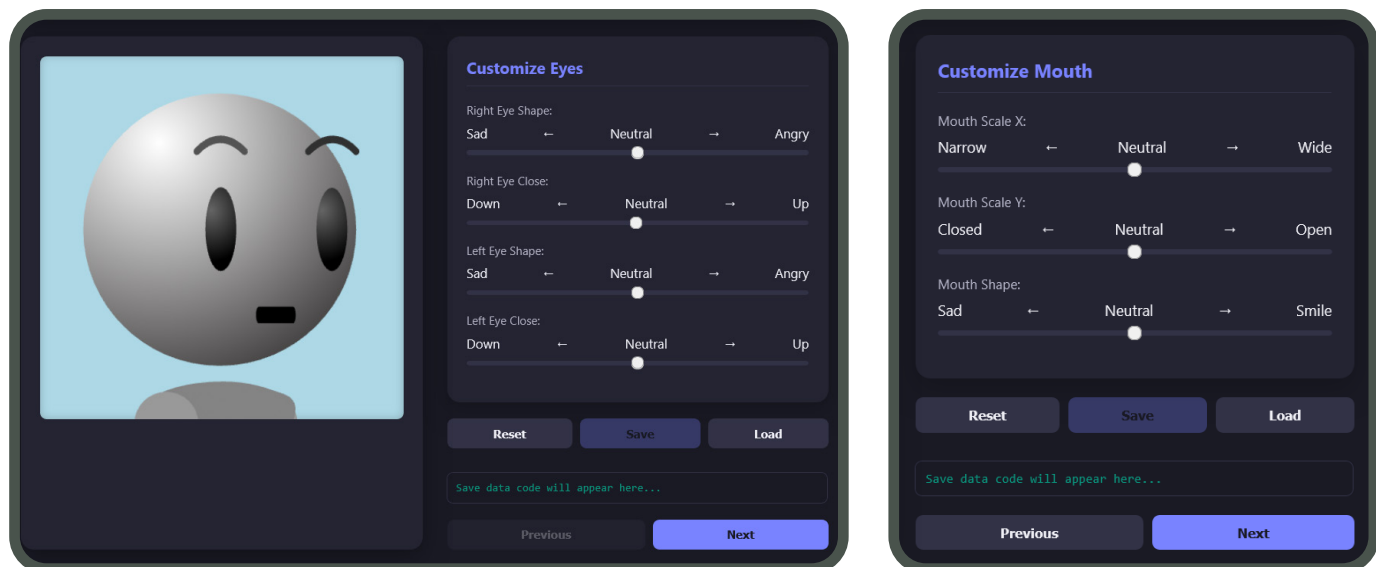
- Examine cultural differences of user-generated visualizations,
- Explore other design strategies (e.g., visual metaphors),
- And expand Data Characters' abilities to represent both affective and contextual information (e.g., life events) in journaling.

Journaling

It was a pretty cool stream, and then we went to the palace, we got some

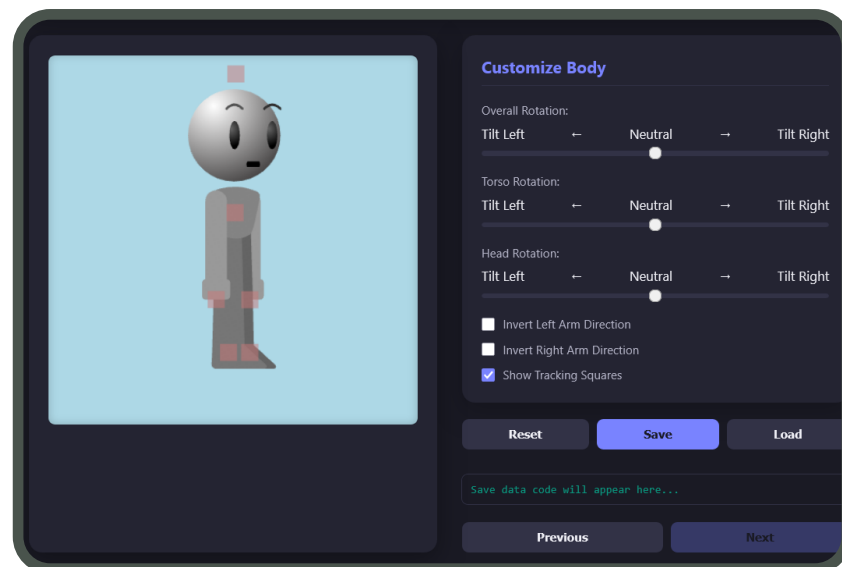
Users reflect on an affective experience.

Facial Customization



Users modify the eyes and mouth regions vital for emotional recognition [7].

Body Customization



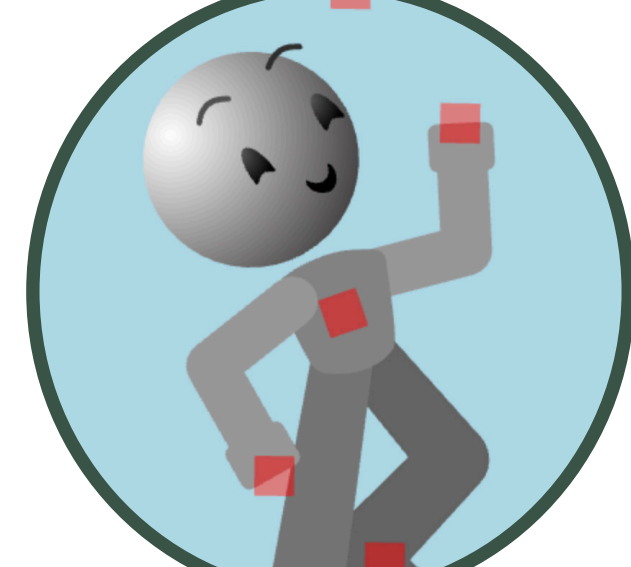
Users modify head, posture, and limb cues [1].

Data Storage



Users' changes save as JSON for future analysis.

Data Character



Final design summarizes affective data.

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